CSC1003 2022 23 Midterm-v2

Computer Science (香港中文大學（深圳））

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CUHK(SZ)-CSC1003 Midterm Exam

Oct. 30, 2022

Name:

Student ID:

Note:

a. This is a closed exam. No notes, calculators, or dictionaries are allowed.
b. This exam paper has five pages, in single-sided printing.
c. Answer all questions within 100 minutes.

Example Question:

Read the following java program, and write down your answer(s).

```java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

What is the output of the program when running "java HelloWorld"?

Hello World!

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1. (10 points) Read the following java program, and write down your answer(s).

```
public class TestNumber1 {
    public static void main(String[] args) {
        int a0 = 0;
        int a1 = 1;
        int a2 = a0 + a1;=1
        int a3 = a1 + a2;=2
        int a4 = a2 + a3;=3
        System.out.println(a0);
        System.out.println(a1);
        System.out.println(a2);
        System.out.println(a3);
        System.out.println(a4);
    }
}
```

What is the command to compile the program?

5' javac TestNumber1.java

What is the output of the program when running "java TestNumber1"?

0 — 1'
1
1
2
3

2. (10 points) Read the following java program, and write down your answer(s).

```
public class TestNumber2 {
    public static void main(String[] args) {
        int a = 3;
        int b = 4;
        a = a + b;
        b = a - b;
        a = a - b;
        System.out.println("a = " + a);
        System.out.println("b = " + b);
    }
}
```

a b
3 4
7 3
4

What is the output of the program when running "java TestNumber2"?

a=4
b=3 — 5'

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3. (15 points) Read the following java program, and write down your answer(s).

```java
public class TestNumber3 {
    public static void main(String[] args) {
        int num1 = Integer.parseInt(args[0]);
        int num2 = Integer.parseInt(args[1]);
        for (int i = num1; i <= num2; i++) {
            int count = 0;
            for (int j = 2; j <= i/2; j++)
                if (i % j == 0) count++;
            if (count == 0) System.out.print(i + " ");
        }
    }
}
```

i = 9

2 9%2 ≠ 0

3 9%3 = 0 count = 1

i ~ [3, 20]

prime 3 i=3 j=2

i=4

3%2

9 2, 3, 4

What is the output of the program when running "java TestNumber3 3 20"?
(Hint: in java, "%" calculates the remainder of integer divisions, i.e., 9 % 3 is 0, 8 % 3 is 2, 7 % 3 is 1, ...)

3 5 7 11 13 17 19

4. (15 points) Read the following java program, and write down your answer(s).

```java
public class TestString {
    public static void main(String[] args) {
        int N = Integer.parseInt(args[0]);
        String ruler = " ";
        for (int i = 1; i <= N; i++) ruler = ruler + i + ruler;
        System.out.println(ruler);
    }
}
```

What is the output of the program when running "java TestString 4"?

1 2 1 3 1 2 1 4 1 2 1 3 1 2 1

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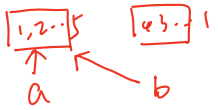
5. (15 points) Read the following java program, and write down your answer(s).

```java
public class TestArray1 {
    public static void main(String[] args) {
        int[] a = {1, 2, 3, 4, 5};
        int[] b = {4, 3, 2, 1};
        System.out.println("a.length = " + a.length);
        System.out.println("b.length = " + b.length);
        b = a; ← After this, b and a refer to the same array
        for (int i = 0; i < b.length; i++) b[i]++;
        System.out.print("Array a:");
        for (int i = 0; i < a.length; i++) System.out.print(" " + a[i]);
        System.out.println();
        System.out.print("Array b:");
        for (int i = 0; i < b.length; i++) System.out.print(" " + b[i]);
    }
}
```



What is the output of the program when running "java TestArray1"?

a.length = 5
b.length = 4 } 5'
Array a: 2 3 4 5 6 5'
Array b: 2 3 4 5 6 5'

6. (15 points) Read the following java program, and write down your answer(s).

```java
public class TestArray2 {
    public static void main(String[] args) {
        int[] a = new int[args.length];
        for (int i = 0; i < args.length; i++)
            a[i] = Integer.parseInt(args[i]);
        for (int i = 0; i < a.length; i++)
            for (int j = i+1; j < a.length; j++)
                if (a[i] > a[j]) {
                    int temp = a[i];
                    a[i] = a[j];
                    a[j] = temp;
                }
        for (int i = 0; i < a.length; i++) System.out.print(a[i] + " ");
    }
}
```

What is the output of the program when running "java TestArray2 9 8 7 6 5 4 3 2 1"?

1 2 3 4 5 6 7 8 9

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7. (5 points) List five or more primitive data types in java language.

int . double . float . char . byte . boolean . long . short

× Integer String × → class

8. (5 points) List two or more types of loops in java language, each with example code.

for (int i=0; i<n; i++) a[i]++;

while (i<n) i++;

do { i++; } while (i<n)

String str1 = "Hello World!"
0 1 2 3 4 5

char ch1 = str1.charAt(5);

9. (5 points) For a two dimensional java array created by "int[][] a = new int[10][5]", what are the values of "a.length" and "a[0].length" respectively?

a.length = 10 ← 50 × a. 10 elements

a[0].length = 5 a = {a[0], a[1], ... a[9]} 10

a[0] = {a[0][1], ... a[0][4]} 5

10. (5 points) What is the usage of "piping" of input/output when running programs?

→ The output of one program is used as the input of the other program

× threads

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